Backend System Business Rules for Admin, Production Support, and Customer Support

1. ***Admin System for Configurable Values***

The configurable values can be set to be effective by a preset date and time. This allows the changes to be scheduled ahead, ensuring smooth transitions and planned updates. Example: An admin sets a new token allocation policy to be effective from the start of the next quarter. This ensures users are notified in advance, and the transition happens seamlessly.

1.1 Configurable Values:

- **Token Allocation:** Admins can configure token values in round numbers (e.g., 1-10 tokens) for:
- Initial token allocation during onboarding.
- Token amounts for hosting, seeking, and other transactions. Example: An admin may set the onboarding token to 5 tokens and seeking a parking match to 2 tokens.
- **Hosting Duration:** Configurable in minutes (e.g., 5-60 minutes). Example: Admins might set the default hosting duration to 30 minutes with flexibility for adjustment based on user needs.

1.2 Access and Approval Mechanism:

- **Maker-Checker Workflow:**
- A "maker" proposes changes to configuration values.
- A "checker" reviews and approves or rejects the proposed changes.
- Changes only take effect upon approval by the checker.
- **Audit Logs:** Detailed logs for each configuration change, capturing:
- User ID of maker and checker.
- Timestamp of proposal and approval.
- Old and new values.

1.3 Notifications:

- **Automatic Notifications for Configuration Updates:**
- When configuration updates are made, the system sends a notification to all members of the admin team. Example: "Token allocation settings updated from 5 to 10 tokens by Admin123 at

10:30 AM."

- **Rejected Changes:**
- If a configuration update is rejected, a detailed notification is sent specifying the reason. Example: "Change to hosting duration from 30 to 90 minutes rejected by Checker123 due to exceeding maximum allowed duration."

2. ***Production Support for Issue Investigation***

2.1 Access to Logs and Metrics:

- **Geolocation and Timestamp Logs:**
- Viewable for specific matches to verify user arrival and hosting compliance. Example: For a match ID #12345, logs indicate Seeker123 arrived at 2:15 PM while Giver123 vacated the spot at 2:30 PM.
- **User Interaction Logs:**
- Access logs of preset messages and other in-app actions during matches. Example: Preset message "I have arrived" sent by Seeker123 at 2:14 PM.

2.2 Investigation Tools:

- **Transaction Summaries:**
- Includes detailed records of all transactions associated with matches, such as token exchanges, timestamps, and involved parties. Example: "Match ID #12345: Seeker123 paid 2 tokens, hosting started at 2:00 PM, completed at 2:30 PM."
- **Match Details:**
- Comprehensive information about active and completed matches, including geolocation data, user interactions, and match outcomes. Example: "Match ID #56789: Hosted by Giver123, seeker arrived at 2:14 PM, swap confirmed at 2:20 PM."
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- **Dispute Details Dashboard:**
- Displays all disputes flagged for investigation.
- Includes transaction summaries, match statuses, and related user activities. Example: Dispute #56789 shows a mismatch in hosting duration claimed by Giver123 and verified geolocation data.
- **Metrics Reports:**

- Summaries of frequent dispute reasons, average resolution times, and unresolved cases. Example: "Frequent disputes: Seeker No-show (30%), Giver left early (20%)" with an average resolution time of 48 hours.

2.3 Audit and Activity Logs:

- Maintain logs for all actions taken by production support, including:
- Dispute flag reviews. Example: "Dispute #56789 flagged by SupportAgent321 for manual review."
- Changes to match statuses or token refunds. Example: "Match ID #12345 status changed to 'Resolved' with 2 tokens refunded to Seeker123."
- Data accessed during investigations. Example: "SupportAgent321 accessed geolocation logs for Match ID #12345."

2.4 Notifications:

- Alert production support team for high-priority unresolved disputes or anomalies detected in system metrics. Example: "Alert: Dispute #56789 remains unresolved for 72 hours. Immediate review required."

2.5 Backend Monitoring:

- **MVP:** Implement basic monitoring to track load time and processing time in real-time; set alerts for deviations beyond 10%. Examples:
- **Load Time Monitoring:** Monitor the time taken to load key system components such as user dashboards and transaction pages. Example: If dashboard load time exceeds 1 second, an alert is triggered.
- **Processing Time Monitoring:** Track the duration of backend operations such as token transactions, user authentication, and dispute resolution. Example: If token transaction processing exceeds 2 seconds, an alert should be raised to investigate potential system bottlenecks.
- **API Response Time:** Measure response times for critical APIs (e.g., match creation or token transfer). Example: Alert if API response time exceeds 3 seconds for more than 5% of requests within a 10-minute window.

3. ***Customer Support for Dispute Handling***

3.1 Dispute Resolution Workflow:

- **Automated Resolution:**

- System attempts to resolve disputes using geolocation and timestamp logs within 24 hours. Example: If a Seeker claims they arrived on time, the system checks GPS data to verify arrival before the hosting period ends.
- Disputes unresolved by automation are flagged for manual review.
- **Manual Review:**
- Customer support reviews logs, presets, and user-provided evidence, such as photos or messages. Example: If a Giver claims the spot was inaccessible, support reviews communication logs and photos provided by both parties.
- Has override privileges to adjust match outcomes or token transactions. Example: Overriding an automated decision to refund a token if new evidence supports the Seeker.

3.2 Access to Logs:

- Full access to:
- Historical logs for all users. Example: Support can access all previous matches and disputes involving User123.
- Communication records and transaction details. Example: Reviewing preset messages sent during a disputed match.

3.3 Override Privileges:

- Ability to override automated resolutions in exceptional cases with justifications logged.
- Example: Adjusting token refunds if evidence shows the Seeker was blocked by unforeseen construction.
- Example: Escalating to higher-level support teams for disputes involving unclear or incomplete data.

3.4 Escalation Mechanism:

- Disputes unresolved after manual review escalate to:
- **Step 1:** Specialist Team (Production Support) for further investigation. Example: Reviewing system-level logs to identify potential technical errors.
- **Step 2:** Admin Committee for final resolution. Example: Resolving policy-related disputes or those with significant user impact.

3.5 Notifications:

- Notify users of:

- Dispute submission acknowledgment. Example: "Your dispute for Match ID #56789 has been received and is under review."
- Updates during automated or manual review. Example: "Dispute #56789 is now being reviewed by customer support."
- Final resolution outcomes. Example: "Dispute #56789 resolved: 2 tokens refunded to your account."

4. ***General Features for Backend Systems***

4.1 Additional Metrics and Logs:

- **User Activity Metrics:**
- Number of matches and cancellations per user. Example: User123 has 15 successful matches and 2 cancellations in the past month. **Usage:** Identifies user engagement and reliability trends, helping to address frequent cancellations proactively and improve user satisfaction.
- Average hosting and seeking durations. Example: Average hosting duration for User123 is 28 minutes, while average seeking time is 12 minutes. **Usage:** Highlights system efficiency and areas for optimization, such as improving matching algorithms or addressing delays.
- **Dispute Trends:**
- Frequency of disputes per user. Example: User123 has initiated 3 disputes, all related to late arrivals by Givers. **Usage:** Flags potential misuse or dissatisfaction, enabling targeted resolution efforts and policy adjustments.
- Common reasons and resolution times. Example: 40% of disputes are due to "Seeker no-shows," with an average resolution time of 36 hours. **Usage:** Reveals systemic issues and enables refinement of dispute resolution processes and user guidelines.
- **Token Economy Metrics:**
- Tokens earned and spent by users. Example: User123 earned 20 tokens from hosting and spent 18 tokens on seeking matches in the last month. **Usage:** Monitors token flow to maintain a balanced economy and adjust rules for earning or spending tokens if needed.
- Outstanding borrowed tokens and repayment statuses. Example: User123 borrowed 4 tokens, repaid 3, and has 1 token overdue. **Usage:** Tracks financial accountability within the app, preventing abuse of borrowing privileges and ensuring a sustainable token ecosystem.

4.2 Escalation Flow:

1. **Automated Resolution:** System resolves disputes within 24 hours using predefined criteria. Example: Dispute #56789 automatically resolved in favor of Seeker123 based on GPS logs.
2. **Manual Review by Customer Support:** Resolved within an additional 48 hours, if possible. Example: Customer support determines a partial token refund is appropriate after reviewing both parties' logs.
3. **Specialist Team (Production Support):** In-depth analysis for unresolved cases. Example: Specialist team identifies a system error causing a dispute and recommends a full token refund.
4. **Admin Committee:** Final resolution within 72 hours of escalation. Example: Admin committee rules on a policy exception for a unique case involving multiple disputes.

4.3 Data Retention Policies:

- Retain logs for 12 months, including historical data for dispute resolution and trend analysis. Example: Logs of disputes from January 2023 are available until January 2024.
- Retain geolocation and timestamp data for unresolved disputes for an additional 48 hours beyond resolution. Example: GPS data for a resolved dispute on March 1, 2023, will be retained until March 3, 2023.